

AKHIL XAVIER

Embedded Software Engineer

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in akhilxavier



WORK EXPERIENCE

Software Engineer, Senior

Acuity Intelligent Spaces (post acquisition)

Jan 2025- Ongoing Boulder, CO

- Developed LT6711GXE (HDMI to DP) bridge driver on Intel ADL platform (6.6 kernel).
- Development and customization on i915 DRM sub-system.
- Multiple development, validation, test activities on new QSYS NVM device.

Embedded Software Engineer

QSC LLC (acquired by Acuity Intelligent Spaces)

May 2023 - Dec 2024 Boulder, CO

- Enabled hardware accelerated QSV codec stack on Intel ADL platform.
- Optimized and validated hardware accelerated encode-decode performance, GPU performance, video post processing (VPP) performance on intel ADL platform.
- Bring up of new encoder-decoder hardware units for a video streaming project.

Software Engineer

Mersive Technologies Inc

Nov 2019 - March 2023 Denver, CO

(Nov 2019 - Dec 2022 India)

- Developed end-to-end HDMI input stack (TMDS-to-MIPI CSI), including driver, framework uEvent listener, and pre-view application.
- Implemented MI2S audio on Qualcomm SoC platforms.
- Improved CPU performance and thermal efficiency.
- Brought up multiple Mersive Pod hardware platforms.

Senior Software Engineer

Inforce Computing India Pvt Ltd

July 2015 - Oct 2019 Bangalore, India

- Developed custom drivers for Bayer/YUV cameras, displays, chargers, HDMI-to-MIPI CSI, and DSI-to-HDMI ICs.
- Customized Qualcomm camera pipelines.
- QCOM SoC-based SBC/SOM bring-up and BSP development.

Software Intern Engineer

Inforce Computing India Pvt Ltd

Nov 2014 - July 2015 Bangalore, India

- Qualcomm SOC based SBC,SOM bringup and BSP.

SKILLS AND EXPERTISE

- Languages: C, C++, Java (Intermediate Level)
- Android OS, Linux distributions (Linaro), Yocto Project
- GStreamer, ffmpeg
- HW Standards: MIPI CSI-2, MIPI-DSI, PCIe, HDMI, I2S, USB, CCI, DDC, I2C, SPI, UART
- Analyzing schematics and datasheets
- Chipsets: APQ8064, APQ8084, APQ8096, APQ8016, SDM660, Intel ADL
- Modules: Camera, Audio, Display, Charger, LowSpeed Peripherals, Image Processing
- BSP development, Linux Device Drivers

EDUCATION

M Tech, Sensor System Technology

VIT University, Vellore

2013 - 2015 Tamil Nadu

B.Tech, Electronics & Communication Engineering

Cochin University of Science and Technology, Kerala

2007 - 2011 Cochin

ACHIEVEMENTS

- Received best paper award during International Science and Technology Conference 2014 held at VIT University.
- Best employee of the month, Mersive Technologies.
- Elected as College Senate member.
- The Project submitted for ESNC 2014 conducted by European Union Space Agency was shortlisted to final stage.
- IEEE paper in EESCO "ISBN:978-1-4799-7676-8" on People Monitoring system.

TASKS ACCOMPLISHED

Driver development and bring-up on QSC hardware

- Platform's : QSC NVM
- SOC : Intel ADL N
- OS's : QSYS OS, Linux kernel 6.6, Linux Kernel 5.15
- LT6711GXE HDMI to DP bridge I2C driver developed on QSC NVM device for dual input channels. The DP output flow to a video processing FPGA and then to SOC through PCIe.
- Enabled and validated hardware accelerated encoding, decoding, video post processing. GPU performance validated. Various performance metrics collected.

- Encoding decoding performance on multiple pixel formats P010LE, YUV422, NV12, YUV444.
- Metrics on thermal performance and power consumption done.
- Enabled device mode operation on USB-C.
- Enabled native out HDMI HDCP support on Linux Intel ADL through libdrm.
- Bring up of various low speed peripherals done.

Multiple porting and customization of Mersive Pods

- Platform's : MER6641
- SOC : APQ8084, APQ8096
- OS's : Lollipop, Nougat, Oreo, Pi, Debian Linaro, Yocto build
- The Mersive pod is ported into upstream Android releases which are supported by the SOC Manufacturer. Also ported the Android based Mersive Pod into Debian based Linaro distribution.
- Customized pods peripherals based on customer usecase.

Driver development and Bring-up of various camera modules across multiple Android and Debian platforms.

- Platform's : IFC6540, IFC6501, IFC6640, IFC6601, IFC6560
- SOC : APQ8084, APQ8096, APQ8016, SDM660, SDA845
- Sensor Module : OV5640 (YUV 5MP), OV7251, (Bayer Monochrome VGA), OV5645 (YUV 5MP), IMX135 (Bayer 13MP), IMX230 (Bayer 21MP), OV8856 (Bayer 8MP), IMX290, OV8856, IMX258.
- OS's : Lollipop, Marsh-mellow, Nougat, kernel 3.10 (Debian), Kernel 4.14 (Debian), Oreo PI, kernel 4.4
- Driver development and bring up of different imaging sensors are done. Different YUV, Bayer, Monochrome camera sensors are brought up.
- www.github.com/akhilxavi/debian4.14-imx230driver

Driver development and bring-up of HDMI to CSI convertor IC TC358840 and ADV7782

- Platform's : IFC6540, IFC6501, IFC6640, IFC6601+, Mersive Pod
- SOC & FPGA : APQ8084, APQ8096, TC358840 (4K HDMI to MIPI CSI-2 convertor), ACC1s80
- Camera Layers : Kernel, server process, media-controller, Android HAL, V4L2
- OS's : Lollipop, Nougat
- Implemented HDMI-to-MIPI CSI via TC358840/ADV7782 FPGA, with uEvent-based hot-plug, resolution/FPS detection and dynamic camera pipeline reconfiguration.

Bring up, Feature implementation, BSP creation and bug Fixing across multiple projects and other miscellaneous works.

- Platform's : IFC6540, IFC6501, IFC6640, IFC6601, IFC6309
- SOC & FPGA : APQ8084, APQ8096, APQ8016, TC358840
- ACC and HW : ACC-1S80, Si5347
- OS's : Lollipop, Nougat, Marshmallow, KitKat, Debian 3.10 kernel, Debian 4.14, Oreo

- Bug Fix for High Bit-rate video causing system shutdown.
- Fixed the issue of FD (file descriptor) leak in camera pipeline during resolution switch.
- Implemented device-tree overlay mechanism on 4.14 Debian kernel for using multiple sensors on same CSI port.
- www.github.com/akhilxavi/debian4.14-DTOverlay
- Bug fix for memory leak in camera pipeline.
- Developed CCI/I2C read/write command line utilities to peek-poke camera sensor registers
Communicate via IOCTL.
- www.github.com/akhilxavi/CCI-I2C-R-W-utilities
- Porting of IFC6410 from KitKat to lollipop.
- Xilinx DS890 ASIC based MIPI CSI2 frames generator bring up.
- USB Cameras.

Customization of camera Image Processing pipeline across various Platforms/Projects

- Platform's : IFC6540, IFC6501, IFC6640, IFC6601
- SOC's : APQ8084, APQ8096, APQ8016
- ACC Card : ACC-1S80
- OS's : Lollipop, Nougat
- Customized and debugged camera HAL across platforms, enabling dual/simultaneous camera, high-FPS, Y8, YUV422→YUV420, Bayer raw dump, CSI 4K frame sync, and stitched-frame pipeline; resolved memory and FD leaks.

Bring up of Analog and Digital audio across multiple platforms

- Platform's : IFC6540, IFC6501, IFC6640, IFC6601
- SOC : APQ8084, APQ8096, APQ8016, TC358840
- ACC : ACC-1S80, WCD9330(codec), WCD9320(codec), PCM5100a(DAC)
- OS's : Lollipop, Nougat, Marshmallow
- Brought up analog WCD codec audio for mic/HDMI stereo recording (44.1/48 kHz) and digital MI2S audio via SoC aDSP.
www.github.com/akhilxavi/android-3.18-MI2S-audio

Multiple tasks across various platforms and projects

- Brought up Pico projection device BML050; enabled non-standard HDMI resolutions. (1920×1200, 800×600, 854×480) using custom EDID and PLL configuration.
- Configured GPIOs for PWM, I/O, and expansion-connector use cases.
- Configured PMIC regulators and LDOs for multiple peripherals.
- Enabled portrait-mode full-screen camera preview on TV displays.
- Brought up and verified Electronic Image Stabilization (EIS) using SSC.
- Reviewed release notes, test guides, and factory-support documentation during production.
- Improved Mersive Pod performance and implemented external USB caching.
- Resolved multiple software/hardware issues on Mersive Pod.
- Validated hardware platforms for next-generation Mersive products.